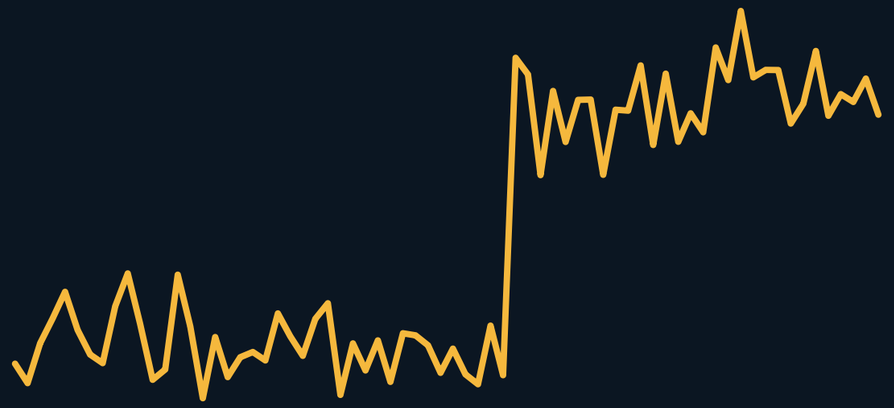
WEEK 3 · DAY 3

The Slow Creep
Drift & CUSUM

Some faults never shout. Today you build a detector that keeps the receipts. 🧾



no single reading looks scary… but something changed

TODAY'S MISSION · 2 HOURS

Three moves



See what 3σ misses

the amplifier's sneaky step



Build CUSUM

add up small leftovers



Watch the slope

a battery ageing faster

⏱️ 10 min hook

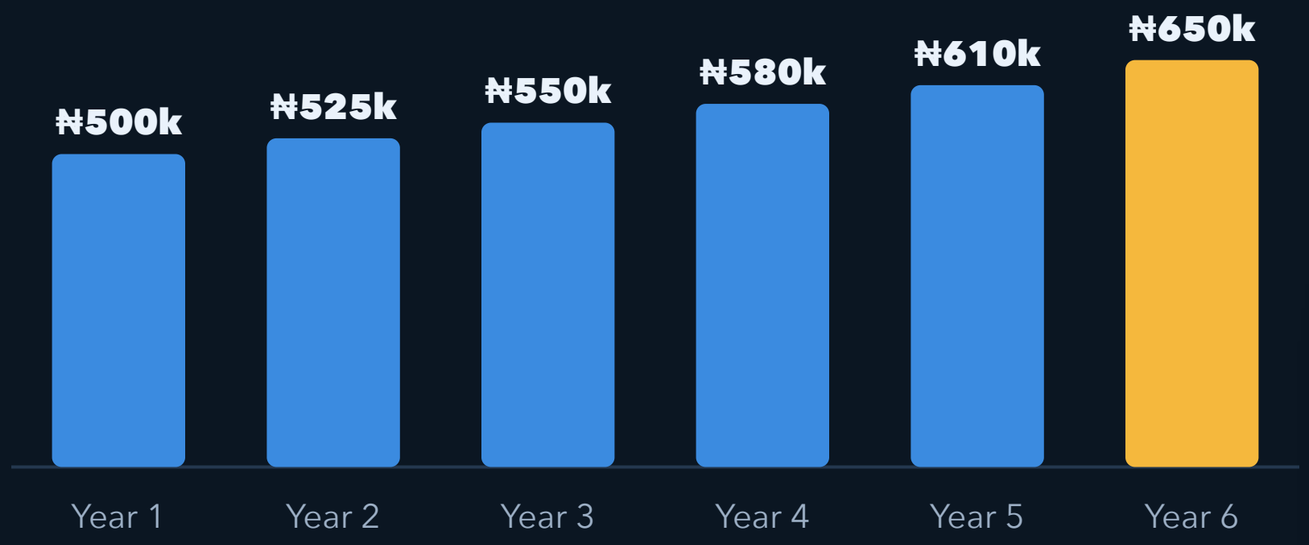
🎤 30 min slides

👩🏾‍💻 60 min hands-on

🏁 20 min wrap-up + project checkpoint

QUICK QUESTION

Did your landlord “increase” the rent? 🏠



illustration, not real rent data 😅

Each year: “small-small” 👈

₦25k here, ₦30k there. No single year shocks you.

Add up the extra: ₦415k

Over 6 years you paid ₦415k more than year 1. **That** is what CUSUM does: it keeps the receipts.

THREE SHAPES OF TROUBLE

Not every fault is a spike

⚡

Spike

one reading way off, then back



🪜

Level shift

a step to a new level that holds



🐌

Drift

a slow lean; the slope changes



Days 1–2 caught **spikes**. Today: the **level shift** and the **drift**.

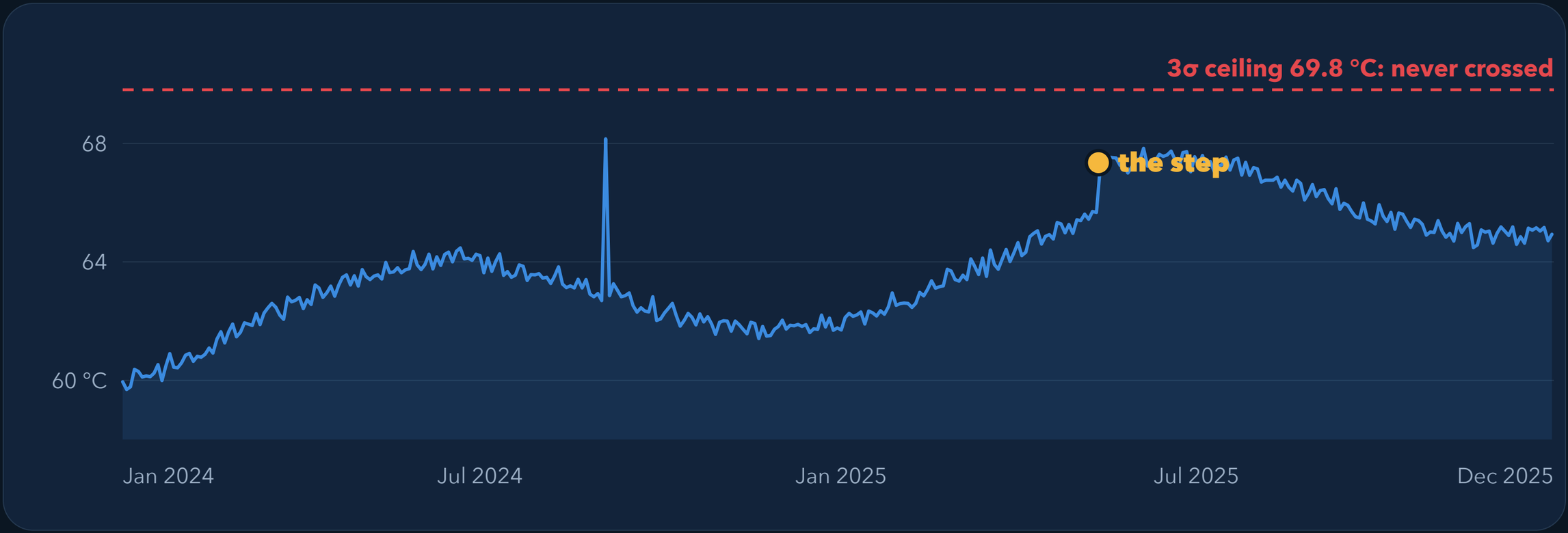
PART 1

The amplifier's sneaky step

🌡️ TWTA: the satellite's loudspeaker. It pushes our signal down to Nigeria, and it gets hot.

TWTA_THERMAL_DAILY.CSV · WEEK-04 VERSION

Two years of amplifier temperature



3σ point rule: 0 alarms

hottest day 68.15 °C (3 Sep 2024)

the step is in there. Can you see it?

ZOOM IN: 30 DAYS BEFORE VS 30 DAYS AFTER

Against its neighbours, it's a giant



+2.17 °C

jump in average

6.9σ

vs the local wobble (0.32 °C)

1.1σ

vs the whole two years (sd 1.97)

STEP 1 · YESTERDAY'S TRICK

Take away what you already know

actual → **− expected**
drive level + sun angle + slow ageing, learned from 2024
→ **leftovers**
residuals



Jan to 11 May 2025: leftovers average -0.01 °C

from 12 May: +1.83 °C, and it stays

STEP 2 · THE CUSUM IDEA

Keep a running total of small leftovers 


```
S = max(0, S + leftover − k)        # every day
```

➕

Add today's leftover

a bit hot? S grows a bit

👈

Minus the slack k

ordinary noise can't pile up

🚨

S above h → alarm

h = how much evidence you need

max(0, ...) means S never goes below zero: good days can't “bank” credit. 🏦❌

DO IT BY HAND · K = 0.5, H = 2.0

Seven days, one receipt book

day	leftover	S
Day 1	+0.3	**0.0**
Day 2	-0.2	**0.0**
Day 3	+0.4	**0.0**
Day 4	+1.0	**0.5**
Day 5	+1.2	**1.2**
Day 6	+0.9	**1.6**
Day 7	+1.1	**2.2** 🚨

The change started on day 4

the day after S was last zero

The alarm rang on day 7

Alarm date ≠ start date. CUSUM needs time to collect evidence. ⏳

THE TWO KNOBS

k and h: set them from σ



k = 0.5 × σ: the slack

Ignore wobble smaller than half a σ. Ordinary noise does **not** accumulate.

Our amplifier: σ = 0.20 °C → **k = 0.10 °C**



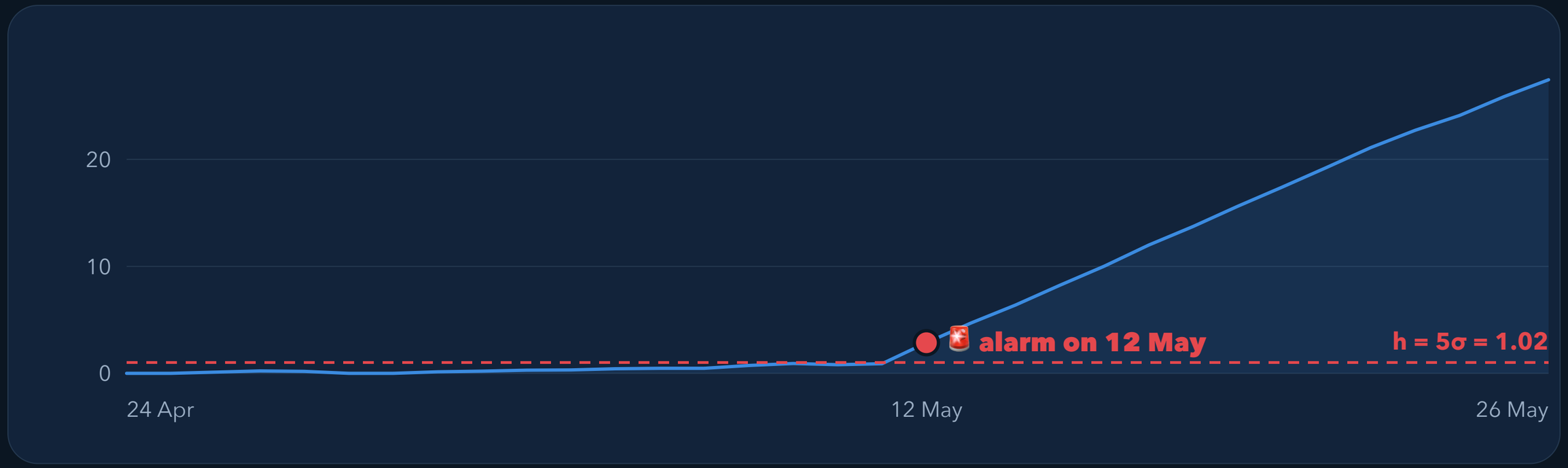
h = 5 × σ: the decision limit

How much evidence before you shout. Low h = fast but noisy. High h = calm but late.

Our amplifier: **h = 1.02 °C**

CUSUM ON THE REAL AMPLIFIER · WATCHING FROM JAN 2025

The receipts pile up on day one



S on 11 May: 0.90 (below h)

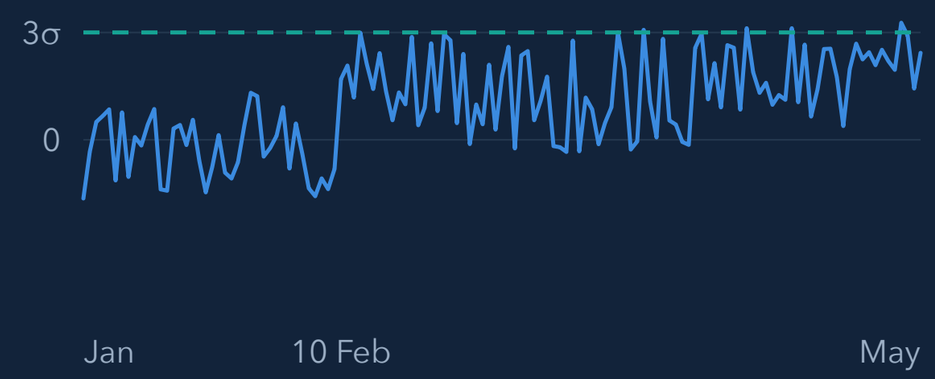
S on 12 May: 2.86 🚨

delay: 0 days, because this step is big (9σ of leftovers)

PRACTICE FAULT: WE ADD +0.3 °C ON PURPOSE

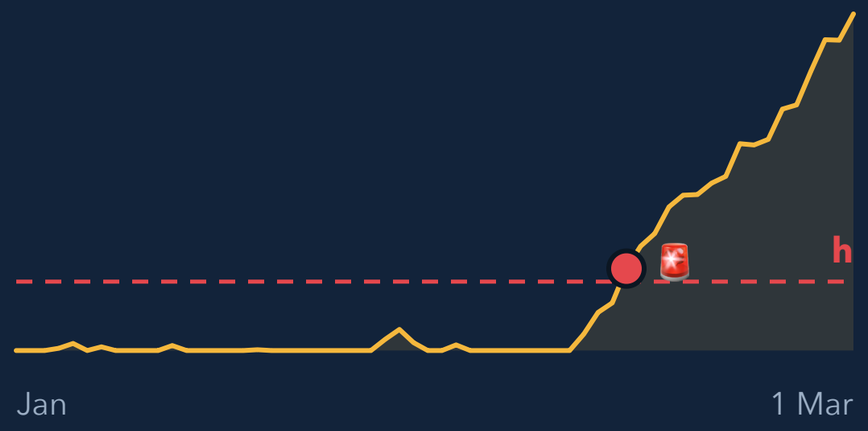
A small fault the point rule never sees

👀 Leftovers vs the 3σ line



0 hits in the 30 days after the fault

🧾 CUSUM



alarm **13 Feb 2025**, 3 days after it began

Where did it start? S was last zero on 9 Feb → change began **10 Feb 2025** ✅

CHOOSING H · SAME PRACTICE FAULT, 404 CALM DAYS

Fast and noisy, or calm and late?

h	false alarms	caught after
2σ	10	1 day
3σ	1	2 days
5σ	0	3 days
8σ	0	5 days
10σ	0	6 days

⚖️ Your choice depends on the cost

Missing a hot amplifier = expensive. A wasted call-out = annoying. Pick h on purpose, and write it down.

🔄 After an alarm: reset

Once someone has checked it, set S back to 0 (or re-learn “normal”). Otherwise CUSUM shouts every day.

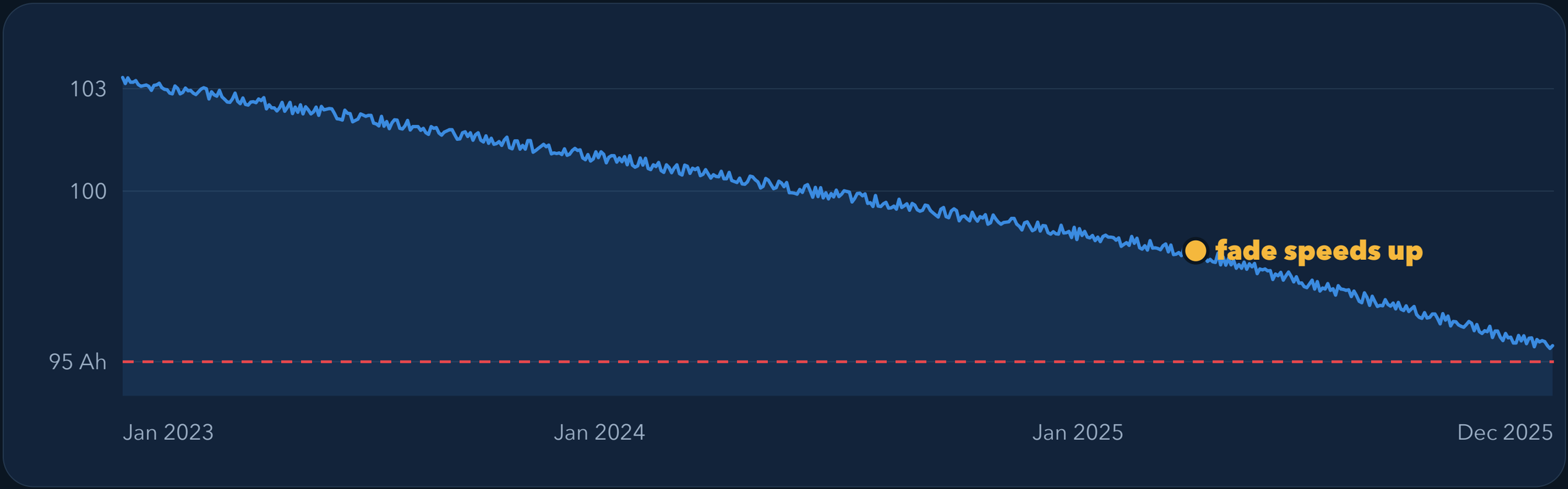
PART 2

The battery that's ageing faster

🔋 The satellite's inverter battery: it carries us through every eclipse.

BATTERY_HEALTH_DAILY.CSV · WEEK-04 VERSION

Three years of usable capacity



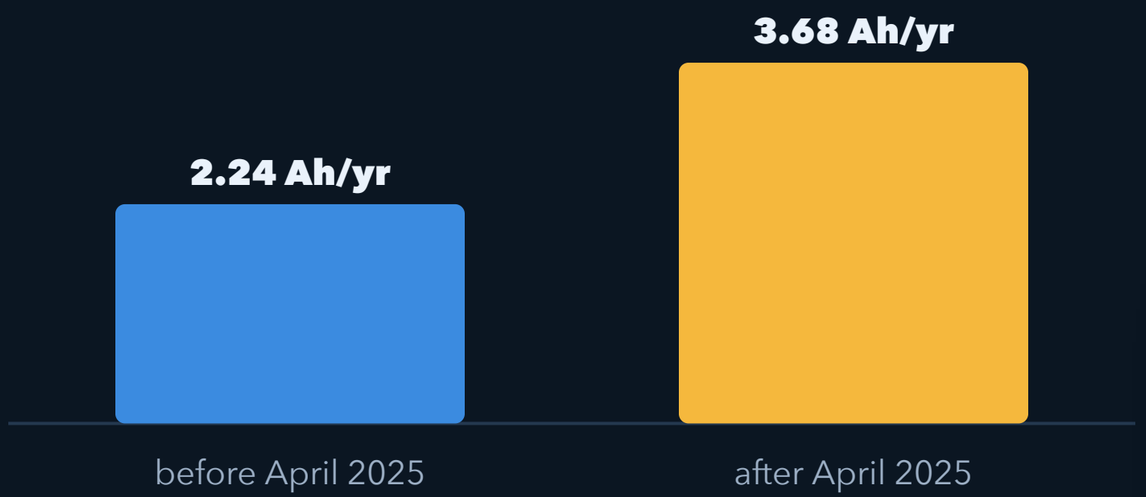
103.3 → 95.4 Ah

95 Ah alarm: 0 days below

⚠ week-03 twin file: no speed-up

THE ANOMALY IS IN THE SLOPE

Not lower. Faster. 🏃🏾



Ah lost per year

a straight line through each period, × 365 days

1.6× faster after April 2025

No single day is unusual. The **speed** changed.

WATCH THE SPEEDOMETER, NOT THE FUEL GAUGE

90-day rolling slope (trailing)



red line: −3.0 Ah/yr warning

before Apr 2025: -2.58 to -1.82

crosses on 2 May 2025 (+31 days)

YOUR TOOLBOX SO FAR

Right tool for each shape 🧰

⚡

Spike

point rule / control chart on the leftovers

Days 1–2



Level shift

CUSUM on the leftovers

today

🐌

Drift / speed-up

rolling slope vs a warning line

today

Always first: **take away what you already know** (season, drive level, eclipse).

🏁 FINAL PROJECT · CHECKPOINT 2

📥 Data in hand

Bring today

✅ your unit head's reply (the email)
✅ your data loaded with a DatetimeIndex + `.asfreq()`
✅ profiled: rows, range, gaps, duplicates
✅ one honest plot with the threshold drawn

Data rules

⛔ real NIGCOMSAT data stays on NIGCOMSAT systems (training server / work laptop), **not Colab**
⛔ no raw data on GitHub
⛔ describe the shape to AI tools, don't paste it

Next checkpoint: **Week 4 • Day 3: method drafted** 🛠️

YOUR TURN · 60 MINUTES

Open the exercise notebook 👩🏾‍💻

- **A** CUSUM by hand: write the one line
- **B** Catch the amplifier's step + a secret fault
- **C** Find when the battery started ageing faster
- 🌶️ Bonus: how noisy is a jumpy h?



14 checks to pass

File → Save a copy in Drive first
Run STEP 0, then top to bottom

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know



A level shift can be huge locally yet invisible to a whole-series 3σ rule.



CUSUM: S = max(0, S + leftover − k). k = slack, h = evidence. Alarm date ≠ start date.



Drift: watch the rolling **slope**, not the level.

Next time 👉 **Scoring a Detector + Quiz 2**: the answer key comes out. 🔑